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COURSE CODE: 17YCM504

COURSE NAME: NATURAL LANGUAGE PROCESSING

UNIT-1 SYLLABUS

LECTURE: 3HRS/WEEK

MARKS: 100

OBJECTIVES:

On completion of the course, student will be able to–	
1	Apply optimization techniques for the improvement of system performance.
2	Ability to understand the synchronous and asynchronous communication mechanisms in their respective OS.
3	Learn about minimization of turnaround time, waiting time and response time and also maximization of throughput with keeping CPU as busy as possible.
4	Ability to compare the different OS
5	Investigate machine translation techniques to facilitate the automatic translation of text between different languages

OUTCOMES

On completion of the course, student will be able to–	
1	Apply optimization techniques for the improvement of system performance.
2	Ability to understand the synchronous and asynchronous communication mechanisms in their respective OS.
3	Learn about minimization of turnaround time, waiting time and response time and also maximization of throughput with keeping CPU as busy as possible.
4	Ability to compare the different OS
5	Ability to understand the concept of Morphology, WSD etc.

Prepared By: Dr. Prajakta Shirke

TEXTBOOKS

Text Books
1. Kosko, B, "Neural Networks and Fuzzy Systems: A Dynamical Approach to Machine Intelligence", Prentice Hall, New Delhi, 2004.
Reference Books
1. Jack M. Zurada, "Introduction to Artificial Neural Systems", PWS Publishing Co., Boston, 2002. 2. Klir G.J. andFolger T.A., "Fuzzysets, Uncertainty and Information", Prentice–Hall of India Pvt. Ltd., New Delhi, 2008. 3. Zimmerman H.J., "Fuzzy set theory and its Applications", Kluwer Academic Publishers Dordrecht, 2001. 4. Driankov,Hellendroonb, "Introduction to fuzzy control", Narosa Publishers, 2001. 5. Laurance Fausett, Englewood cliffs, N.J., "Fundamentals of Neural Networks", Pearson Education, New Delhi, 2008.

SYLLABUS:

Module 1: Natural Language Processing: Introduction to Natural Language Processing Types of NLP systems, How computer understands text, Terminologies used in NLP

Module 2: Steps Involved in NLP, Steps involved in pre-processing, Pipeline of NLP Problems, Challenges in NLP

Prepared By: Dr. Prajakta Shinde

MODULE 1

NATURAL LANGUAGE PROCESSING

WHAT IS NATURAL LANGUAGE?

Natural language refers to the languages humans use for communication, like English, German, or Spanish, with complex rules, grammar, and context developed naturally over time. Unlike formal or programming languages, natural languages are inherently flexible, ambiguous, and rich in nuance, enabling expression of ideas, emotions, and culture.

INTRODUCTION TO NATURAL LANGUAGE PROCESSING

Natural Language Processing (NLP) is one of the hottest areas of artificial intelligence (AI) thanks to applications like text generators that compose coherent essays, chatbots that fool people into thinking they're sentient, and text-to-image programs that produce photorealistic images of anything you can describe. Recent years have brought a revolution in the ability of computers to understand human languages, programming languages, and even biological and chemical sequences, such as DNA and protein structures, that resemble language. The latest AI models are unlocking these areas to analyze the meanings of input text and generate meaningful, expressive output.

Natural language processing (NLP) is the discipline of building machines that can manipulate human language — or data that resembles human language — in the way that it is written, spoken, and organized. It evolved from computational linguistics, which uses computer science to understand the principles of language, but rather than developing theoretical frameworks, NLP is an engineering discipline that seeks to build technology to accomplish useful tasks. NLP can be divided into two overlapping subfields: natural language understanding (NLU), which focuses on semantic analysis or determining the intended meaning of text, and natural language generation (NLG), which focuses on text generation by a machine. NLP is separate from — but often used in conjunction with — speech recognition, which seeks to parse spoken language into words, turning sound into text and vice versa.

NLP is an integral part of everyday life and becoming more so as language technology is applied to diverse fields like retailing (for instance, in customer service chatbots) and medicine (interpreting or summarizing electronic health records). Conversational agents such as Amazon's Alexa and Apple's Siri utilize NLP to listen to user queries and find answers. The most sophisticated such agents — such as GPT-3, which was recently opened for commercial applications — can generate sophisticated prose on a wide variety of topics as well as power chatbots that are capable of holding coherent conversations. Google uses NLP to improve its search engine results, and social networks like Facebook use it to detect and filter hate speech.

NLP is growing increasingly sophisticated, yet much work remains to be done. Current systems are prone to bias and incoherence, and occasionally behave erratically. Despite the challenges, machine learning engineers have many opportunities to apply NLP in ways that are ever more central to a functioning society.

Prepared By: Dr. Prajakta Shirke

HISTORY OF NATURAL LANGUAGE PROCESSING

Natural Language Processing (NLP) is a field of research which determines the way computers can be used to understand and manage natural language text or speech to do useful things. Natural Language Processing (NLP) a subset technique of Artificial Intelligence which is used to narrow the communication gap between the Computer and Human. It is originated from the idea of Machine Translation (MT) which came to existence during the 1950. The primary aim was to translate one human language to another human language, for example, Russian language to English language using the brain of the Computers but after that, the thought of conversion of human language to computer language and vice-versa emerged, so that communication with the machine became easy.

NLP is a process where input provided in a human language and converts this input into a useful form of representation. The field of NLP is primarily concerned with getting computers to perform interesting and useful tasks with human languages. The field of NLP is secondarily concerned with helping us come to a better understanding of human language. As stated above the idea had emerged from the need for Machine Translation in the 1950s. Then the original language was English and Russian. But the use of other words such as Chinese also came into existence in the initial period of the 1960s.

In the 1960s, the NLP got a new life when the idea and need of Artificial Intelligence emerged. In 1978 LUNAR is developed by W.A woods; it could analyze, compare and evaluate the chemical data on a lunar rock and soil composition that was accumulating as a result of Apollo moon missions and can answer the related question. In the 1980s the area of computational grammar became a very active field of research which was linked with the science of reasoning for meaning and considering the user‘s beliefs and intentions. In the period of 1990s, the pace of growth of NLP increased. Grammars, tools and Practical resources related to NLP became available with the parsers. Probabilistic and data-driven models had become quite standard by then. In 2000, Engineers had a large amount of spoken and textual data available for creating systems. Today, large amount of work is being done in the field of NLP using Machine Learning or Deep Neural Networks in general, where we are able to create state-of-the-art models in text classification, Question and Answer generation, Sentiment Classification, etc.

NEED FOR NATURAL LANGUAGE PROCESSING

The growing amount of unstructured natural language data in the world makes it increasingly important for machines to comprehend and analyze it effectively. By training NLP models, we can equip computers to process language data in a variety of forms, from written text to voice input. With centuries of human-written literature and massive data available, it’s vital to teach computers to interpret that wealth of information. However, this task comes with significant challenges, such as resolving ambiguities in meaning, Named-Entity Recognition (NER), and coreference resolution.

While NLP systems are improving, they still face difficulties in understanding the exact meaning of sentences. For example, consider the phrase: "The boy radiated fire-like vibes." Does it refer to a motivating personality or imply something literal? Such ambiguities make text analysis complex for computers.

To solve these challenges, NLP breaks down language understanding into smaller, manageable components. This approach, known as the NLP pipeline, involves several stages that collectively enable machines to interpret human language effective.

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LEVELS OF NLP

There are two components of Natural Language Processing

a. Natural Language Understanding (NLU)

• NLU takes some spoken/typed sentence and working out what it means.

• Here different level of analysis required such as morphological analysis, syntactic analysis, semantic analysis, discourse analysis.

b. Natural Language Generation (NLG)

• NLG takes some formal representation of what you want to say and working out a way to express it in a natural (human) language (e.g., English)

• Here different level of synthesis required: deep planning (what to say), syntactic generation

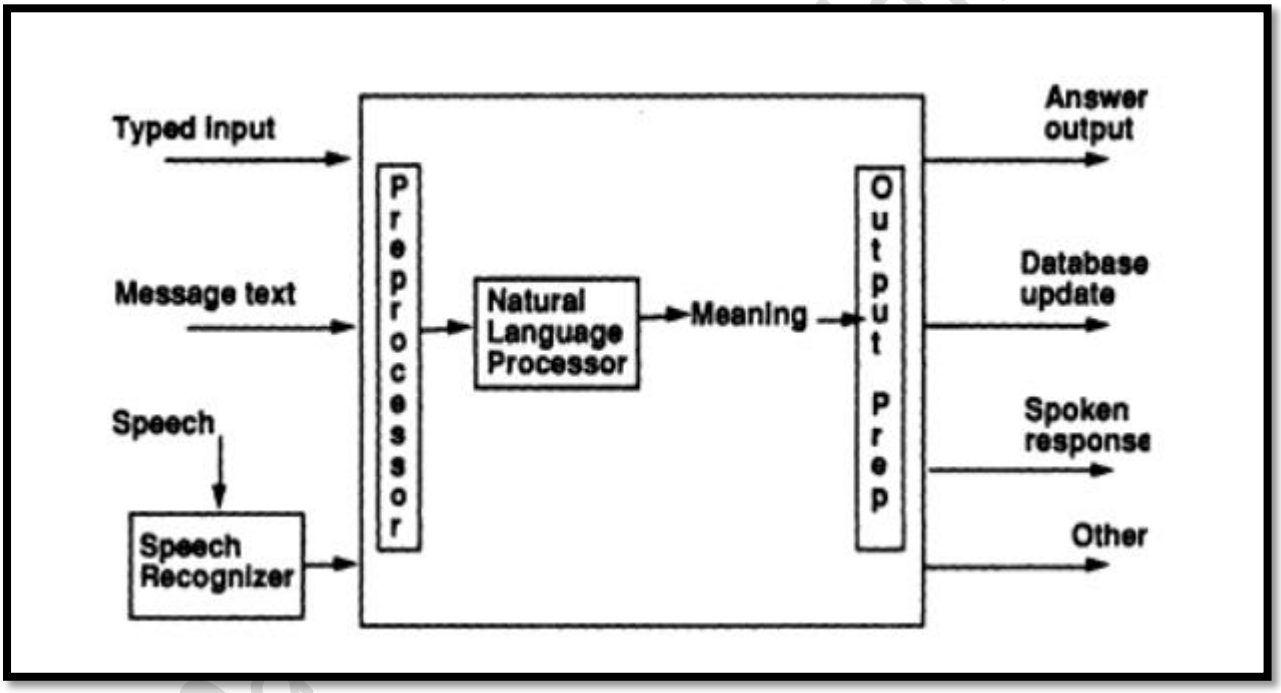

Figure: Generic NLP systems

DIFFERENCE BETWEEN NLU AND NLG

NLU	NLG
It is the process of reading and interpreting language.	It is the process of writing or generating language.
NLU explains the meaning behind the written text or speech in natural language	NLG generates the natural language using machines.
NLU draws facts from the natural language using various tools and technologies such as parsers, POS taggers, etc,	NLG uses the insights generated from parsers, POS tags, etc. to generate the natural language.
NLU understands the human language and converts it into data	NLG uses the structured data and generates meaningful narratives out of it

TYPES OF NLP SYSTEMS

Natural Language Processing (NLP) systems can be categorized based on their functionality, approach, or application. Below are the main types of NLP systems, explained concisely:

1. **Rule-Based Systems**:
   - Use predefined linguistic rules and patterns (e.g., grammar, syntax) to process text.
   - Example: Early chatbots or text parsers that rely on hand-crafted rules.
   - Pros: Precise for specific tasks; no training data needed.
   - Cons: Limited scalability, struggles with ambiguity.
2. **Statistical NLP Systems**:
   - Rely on statistical models and probabilities derived from large datasets to understand language.
   - Example: Word frequency-based models for text classification or translation.
   - Pros: Better at handling variability than rule-based systems.
   - Cons: Requires large datasets; less interpretable.
3. **Machine Learning-Based Systems**:

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- Use machine learning algorithms (e.g., decision trees, SVMs) trained on annotated data to perform tasks like sentiment analysis or named entity recognition.
- Example: Spam email filters or topic modeling tools.
- Pros: Adaptable to new data; improves with more training.
- Cons: Needs labeled data; can be computationally intensive.

4. **Deep Learning-Based Systems**:
   - Employ neural networks (e.g., RNNs, Transformers like BERT) to model complex language patterns.
   - Example: Modern machine translation (e.g., Google Translate), chatbots like me, or voice assistants.
   - Pros: Highly accurate; handles context and ambiguity well.
   - Cons: Requires massive data and computational power; less interpretable.
5. **Hybrid Systems**:
   - Combine rule-based, statistical, and/or deep learning approaches for better performance.
   - Example: Systems that use rules for preprocessing and neural networks for final predictions.
   - Pros: Balances accuracy and efficiency.
   - Cons: Complex to design and maintain.
6. **Task-Specific Systems** (based on application):
   - **Machine Translation**: Translates languages (e.g., German to English, as in your previous query).
   - **Sentiment Analysis**: Determines emotions or opinions in text.
   - **Named Entity Recognition (NER)**: Identifies names, places, or organizations.
   - **Text Summarization**: Generates concise summaries of long texts.
   - **Speech Recognition**: Converts spoken language to text (e.g., Siri).
   - **Question Answering**: Provides answers to user queries (like me!).
   - **Chatbots/Conversational Agents**: Engages in dialogue (e.g., customer service bots).

HOW COMPUTER UNDERSTAND TEXT

Computers understand text through a series of processes that convert human language into a format they can process. Here's a concise explanation:

1. **Text Representation**: Computers convert text into numerical formats since they only understand numbers.
   - **Character Encoding**: Text is encoded using standards like ASCII or Unicode, where each character (e.g., 'A', 'b') is mapped to a unique number.
   - **Tokenization**: Text is broken into smaller units like words or subwords (tokens). For example, "I love coding" becomes ["I", "love", "coding"].
   - **Word Embeddings**: Tokens are converted into vectors (arrays of numbers) using techniques like Word2Vec, GloVe, or transformer-based embeddings (e.g., BERT). These vectors capture semantic meaning, so "cat" and "kitten" have similar vectors.
2. **Natural Language Processing (NLP)**: Computers use algorithms to analyze and interpret text.
   - **Parsing**: The structure of text (e.g., grammar, syntax) is analyzed to understand relationships between words.
   - **Context Analysis**: Models like transformers (used in GPT, BERT) analyze context by looking at surrounding words or sentences to infer meaning.

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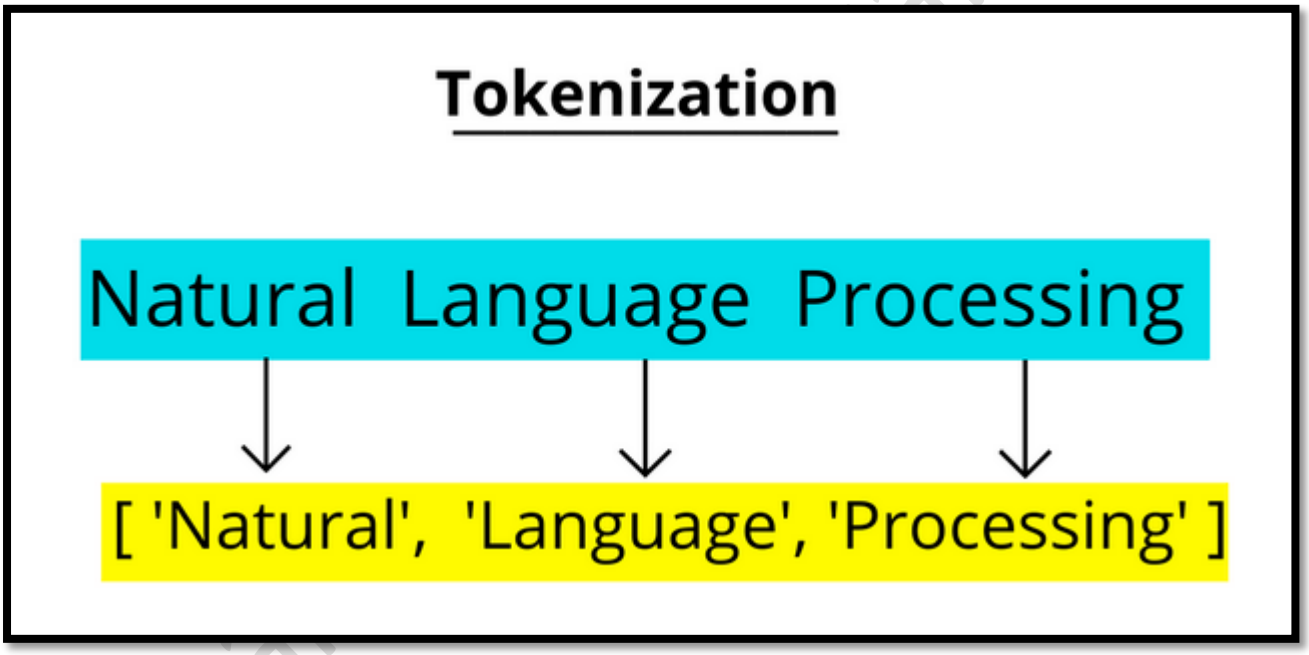
- **Machine Learning Models**: Trained on vast datasets, models learn patterns in language, enabling tasks like sentiment analysis, translation, or question answering.

3. **Applications**: Once processed, computers can perform tasks like:
   - Classifying text (e.g., spam detection).
   - Generating text (e.g., chatbots like me).
   - Translating languages or summarizing content.

TERMINOLOGIES USED IN NLP

Tokenization

Also called word segmentation, tokenization is one of the simplest and most important techniques involved in NLP. It's a crucial pre-processing step in which a long string of text is broken down into smaller units called tokens. Tokens include words, characters, and sub words. They are the building blocks of natural language processing, and most NLP models process raw text on the token level.



Stemming & lemmatization

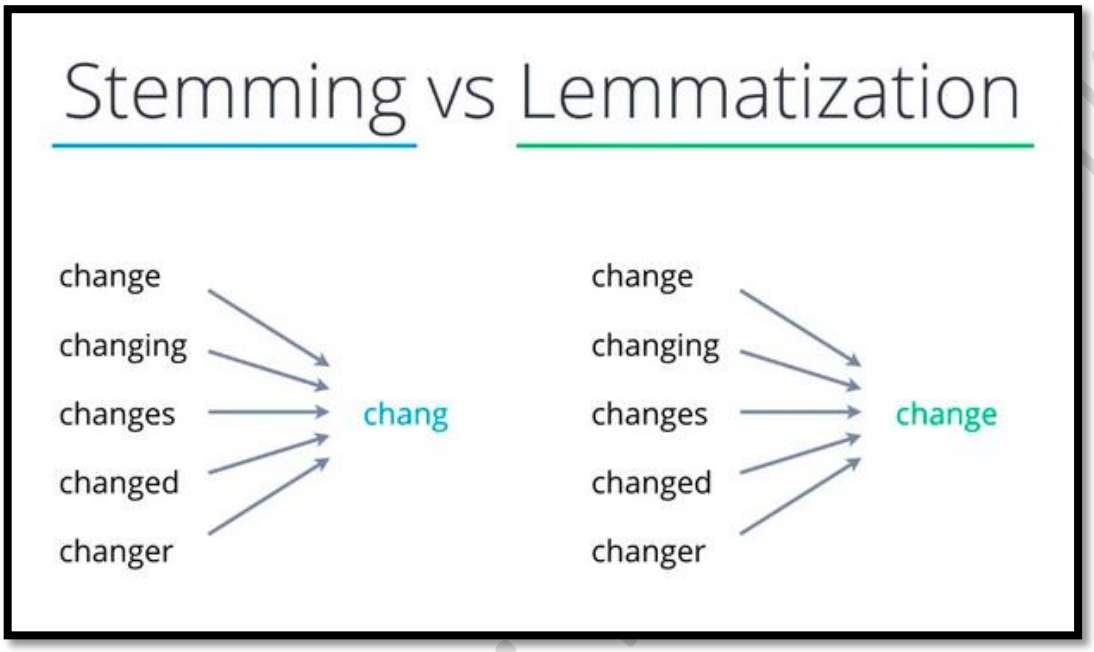
After tokenization, the next preprocessing step is either stemming or lemmatization. These techniques generate the root word from the different existing variations of a word.

For example, the root word "stick" can be written in many different variations, like:

- Stick
- Stuck
- Sticker
- Sticking
- Sticks

- Unstick

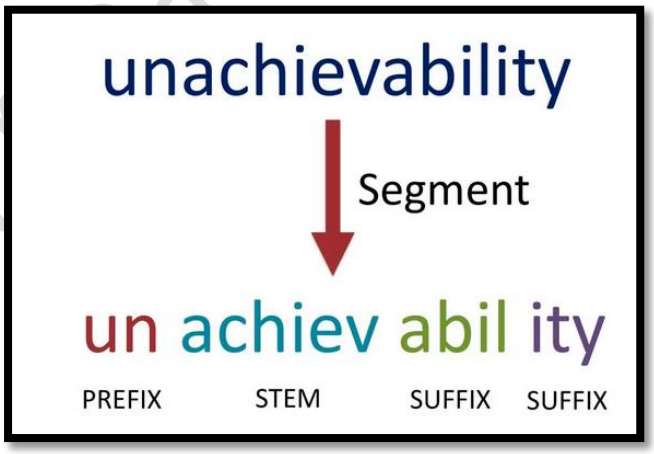
Stemming and lemmatization are two different ways to try to identify a root word. Stemming works by removing the end of a word. This NLP technique may or may not work depending on the word. For example, it would work on "sticks," but not "unstick" or "stuck." Lemmatization is a more sophisticated technique that uses morphological analysis to find the base form of a word, also called a lemma.



Morphological segmentation

Morphological segmentation is the process of splitting words into the morphemes that make them up. A morpheme is the smallest unit of language that carries meaning. Some words such as "table" and "lamp" only contain one morpheme.

But other words can contain multiple morphemes. For example, the word "sunrise" contains two morphemes: sun and rise. Like stemming and lemmatization, morphological segmentation can help preprocess input text.



Stop words removal

Stop words removal is another preprocessing step of NLP that removes filler words to allow the AI to focus on words that hold meaning. This includes conjunctions such as “and” and “because,” as well as prepositions such as “under” and “in.”

By removing these unhelpful words, NLP systems are left with less data to process, allowing them to work more efficiently. It isn’t a necessary step of every NLP use case, but it can help with things such as text classification.

Text classification

Text classification is an umbrella term for any technique used to organize large quantities of raw text data. Sentiment analysis, topic modeling, and keyword extraction are all different types of text classification. And we’ll talk about them shortly. Text classification essentially takes unstructured text data and structures it, preparing it for further analysis. It can be used on nearly every text type and help with a number of different organization and categorization applications.

Some possible applications of text classification include:

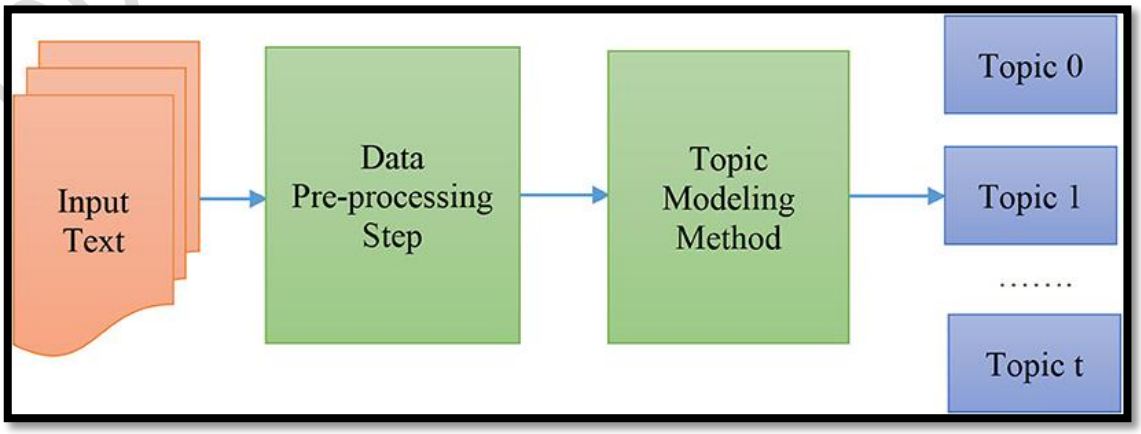
- Grouping product reviews into categories based on sentiment.
- Flagging customer emails as more or less urgent.
- Organizing content by topic.

Sentiment analysis

Sentiment analysis, also known as emotion AI or opinion mining, is the process of analyzing text to determine whether it is generally positive, negative, or neutral. As one of the most important NLP techniques for text classification, sentiment analysis is commonly used for applications such as analysing user-generated content. It can be used on a variety of text types, including reviews, comments, tweets, and articles.

Topic modelling

Topic modeling is a technique that scans documents to find themes and patterns within them, clustering related expressions and word groupings as a way to tag the set. It’s an unsupervised machine learning process, meaning that it doesn’t require the documents it is processing to have previously been categorized by humans.



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Keyword Extraction

Keyword extraction is a technique that skims a document, ignoring the filler words and honing in on the important keywords. It is used to automatically extract the most frequently used and essential words and phrases from a document, helping to summarize it and identify what it's about.

This is highly useful for any situation in which you want to identify a topic of interest in a textual dataset, such as whether there is a problem that comes up again and again in customer emails.

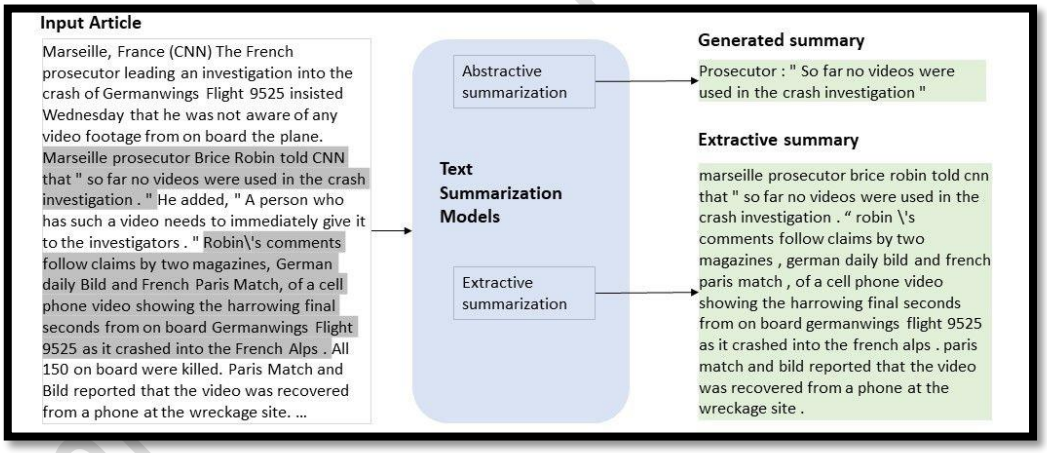
Text summarization

This NLP technique summarizes a text in a coherent way, and it's great for extracting useful information from a source. While a human would have to read an entire document in order to write an accurate summary of it, which takes quite a bit of time, automatic text summarization can do it much more quickly.

There are two types of text summarization:

Extraction-based – This technique pulls key phrases and words from the document to make a summary without changing the original text.

Abstraction-based – This technique creates new phrases and sentences based on the original document, essentially paraphrasing it.



Parsing

Parsing is the process of figuring out the grammatical structure of a sentence, determining which words belong together as phrases and which are the subject or object of a verb. This NLP technique offers additional context about a text in order to help with processing and analyzing it accurately.

Named Entity Recognition

Named entity recognition (NER) is a type of information extraction that locates and tags "named entities" with predefined keywords such as names, locations, dates, events, and more.

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In addition to tagging a document with keywords, NER also keeps track of how many times a named entity is mentioned in a given dataset. NER is similar to keyword extraction, but the extracted keywords are put into predefined categories.

NER can be used to identify how often a certain term or topic is mentioned in a given data set. For example, it might be used to identify that a certain issue, tagged as a word like "slow" or "expensive," comes up again and again in customer reviews.

Prepared By: Dr. Mirke

MODULE 2

STEPS INVOLVED IN NATURAL LANGUAGE PROCESSING

There are general five steps in natural language processing

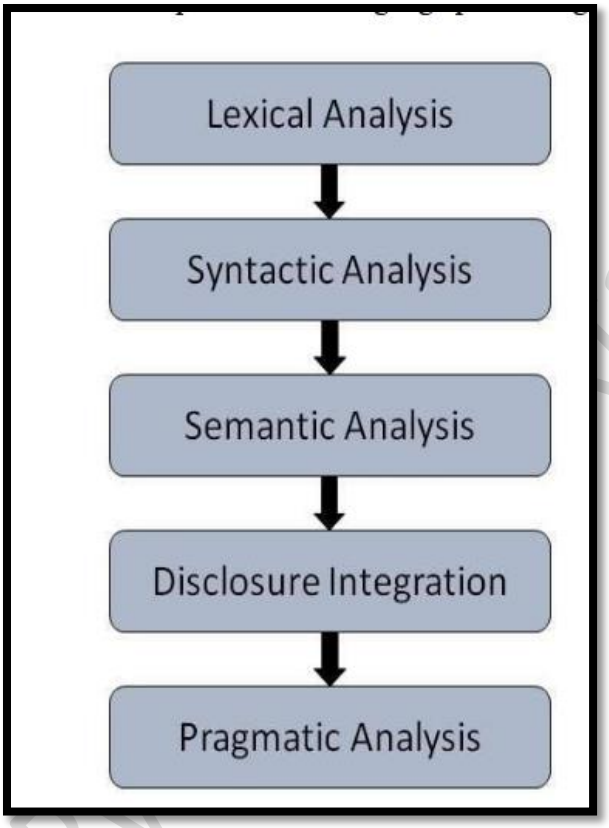

Figure: steps involved in Natural Language Processing

Lexical Analysis:

It is the first stage in NLP. It is also known as morphological analysis. It consists of identifying and analyzing the structure of words. Lexicon of a language means the collection of phrases and words in a language. Lexical analysis is dividing the whole chunk of text into words, sentences, and paragraphs

Syntactic Analysis:

Syntactic analysis consists of analysis of words in the sentence for grammar and ordering words in a way that shows the relationship among the words. For example the sentence such as “The school goes to boy” is rejected by English syntactic analyzer.

Semantic Analysis:

Semantic analysis is a structure created by the syntactic analyser which assigns meanings. This component transfers linear sequences of words into structures. It shows how the words are associated with each other.

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Semantics focuses only on the literal meaning of words, phrases, and sentences. This only draws the dictionary meaning or the real meaning from the given text. The structures assigned by the syntactic analyzer always have assigned meaning. The text is checked for meaningfulness. It is done by mapping syntactic structure and objects in the task domain. E.g. "colorless green idea". This would be rejected by the Symantec analysis as colorless here; green doesn't make any sense.

Discourse Integration:

The meaning of any sentence depends upon the meaning of the sentence just before it. Furthermore, it also brings about the meaning of immediately succeeding sentence. For example, "He wanted that", in this sentence the word "that" depends upon the prior discourse context.

Pragmatic Analysis:

Pragmatic analysis concerned with the overall communicative and social content and its effect on interpretation. It means abstracting or deriving the meaningful use of language in situations. In this analysis, what was said is reinterpreted on what it truly meant. It contains deriving those aspects of language which necessitate real world knowledge. E.g., "close the window?" should be interpreted as a request instead of an order.

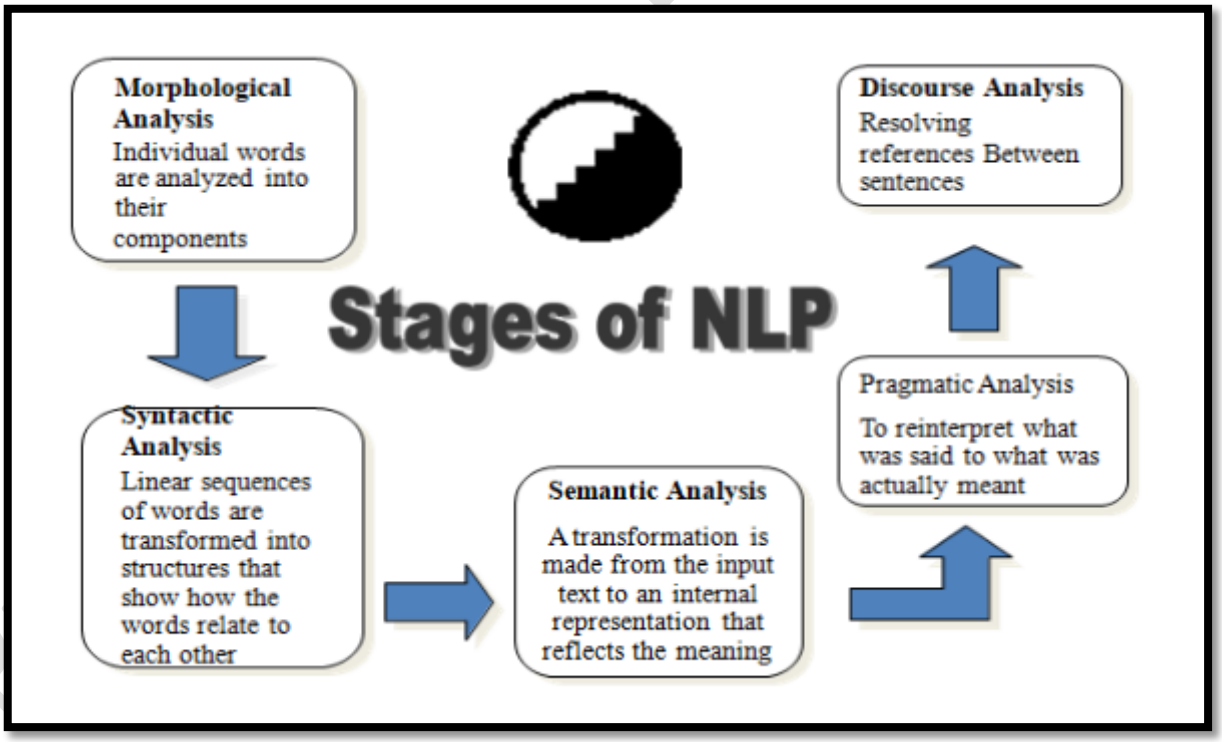


Figure: Stages of NLP

Prajakta Shirke

Morphological Analysis:

Consider we have an English interface to an operating system and the following sentence is typed: "I want to print Bill's .init file ". Morphological analysis must do the following things:

• Pull apart the word "Bill's" into proper noun "Bill" and the possessive suffix "'s".

• Recognize the sequence ".init" as a file extension that is functioning as an adjective in the sentence.

This process will usually assign syntactic categories to all the words in the sentence. Consider the word "prints". This word is either a plural noun or a third person singular verb (he prints)

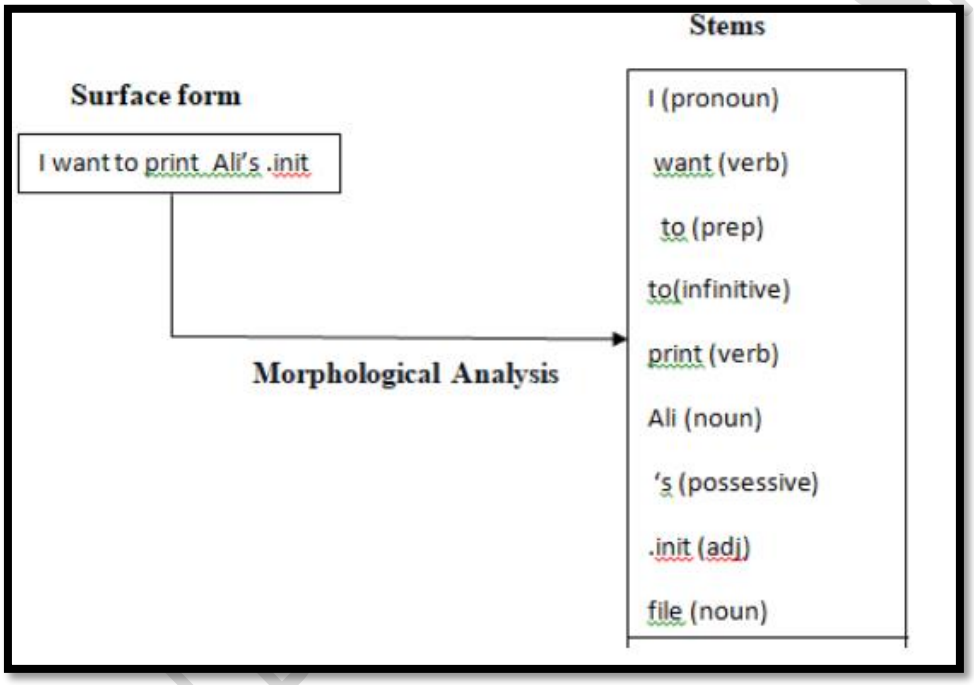


Figure: Morphological Analysis Example

Syntactic analysis:

This method examines the structure of a sentence and performs detailed analysis of the sentence and semantics of the statement. In order to perform this, the system is expected to have through knowledge of the grammar of the language. The basic unit of any language is sentence, made up of group of words, having their own meanings and linked together to present an idea or thought. Apart from having meanings, words fall under categories called parts of speech. In English languages, there are eight different parts of speech. They are nouns, pronoun, adjectives, verb, adverb, prepositions, conjunction and interjections. In English language, a sentence S is made up of a noun phrase (NP) and a verb phrase (VP), i.e.

S=NP+VP

The given noun phrase (NP) normally can have an article or delimiter (D) or an adjective (ADJ) and the noun (N), i.e.

NP=D+ADJ+N

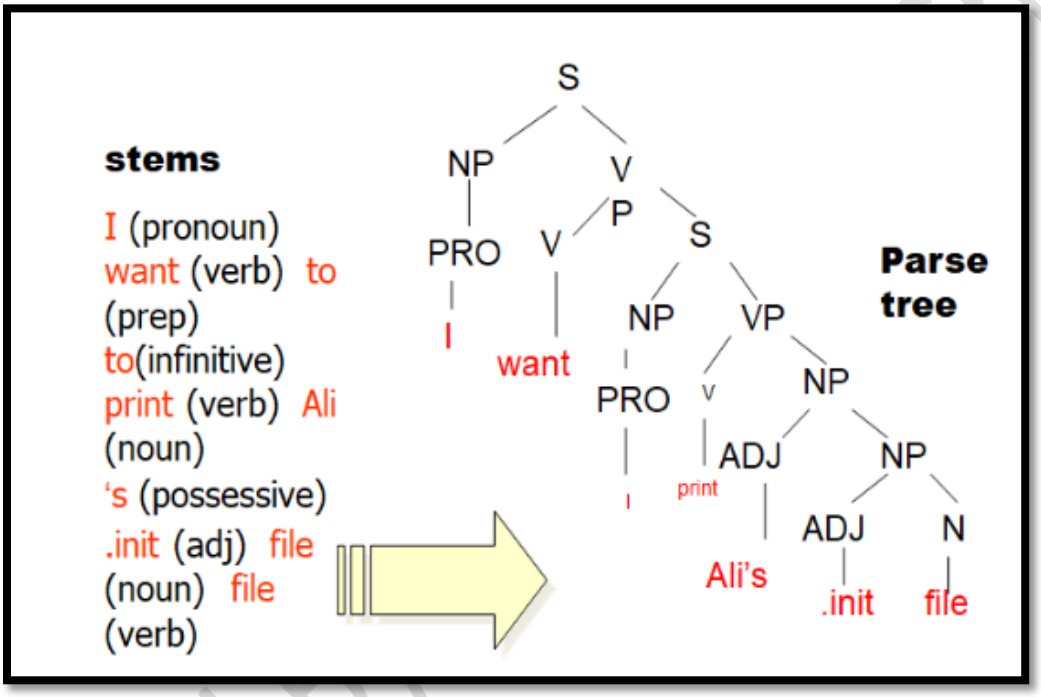
Also a noun phrase may have a prepositional phrase (PP) which has a preposition (P), a delimiter (D) and the noun (N), i.e.

PP=D+P+N

The verb phrase (VP) has a verb (V) and the object of the verb. The object of the verb may be a noun (N) and its determiner, i.e.

VP=V+N+D

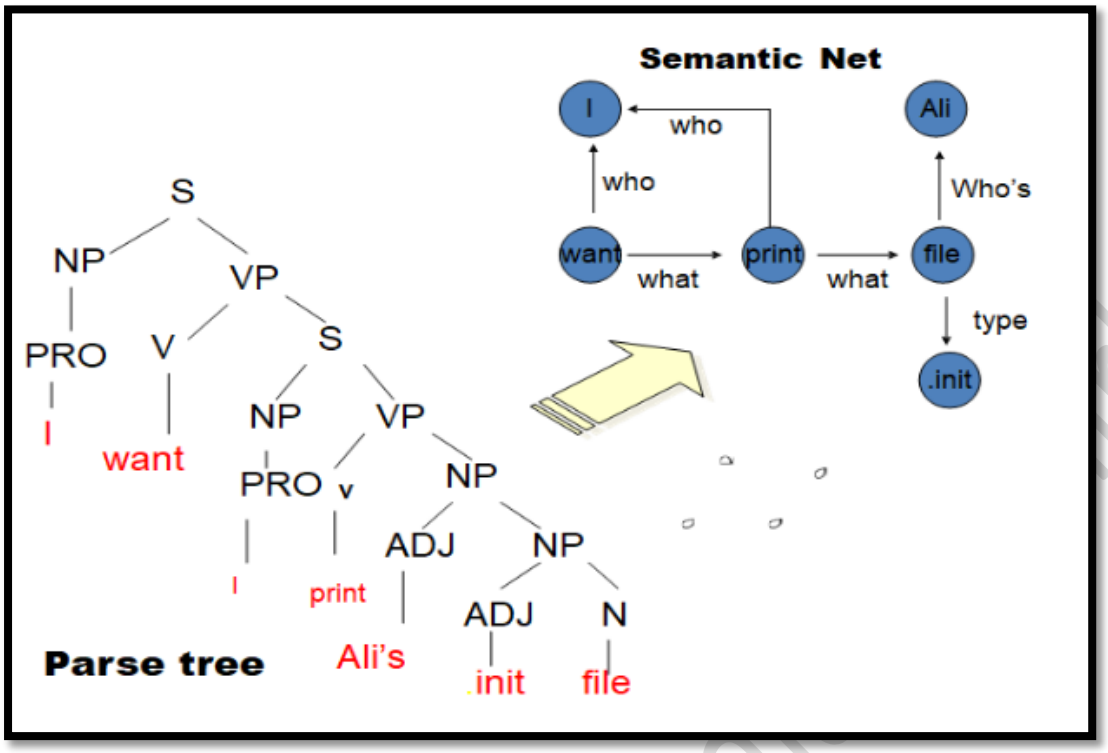
These are some of the rules of the English grammar that helps one to construct a small parser for NLP.



Semantic Analysis:

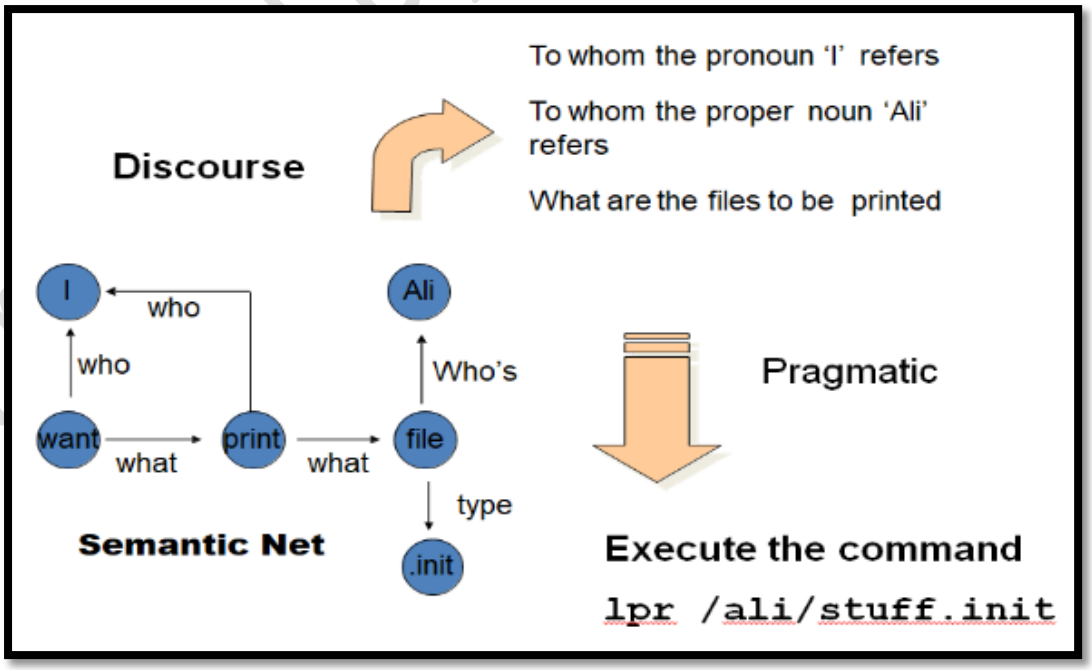
The structures created by the syntactic analyzer assigned meanings. Also, a mapping made between the syntactic structures and objects in the task domain. Moreover, Structures for which no such mapping possible may reject.

Semantic analysis must do two important things: It must map individual words into appropriate objects in the knowledge base or database It must create the correct structures to correspond to the way the meanings of the individual words combine with each other.



Discourse Integration:

The meaning of an individual sentence may depend on the sentences that precede it. And also, may influence the meanings of the sentences that follow it Specifically we do not know whom the pronoun "I" or the proper noun "Bill" refers to. To pin down these references requires an appeal to a model of the current discourse context, from which we can learn that the current user is USER068 and that the only person named "Bill" about whom we could be talking is USER073.Once the correct referent for Bill is known, we can also determine exactly which file is being referred to.



Pragmatic Analysis

Moreover, the structure representing what said reinterpreted to determine what was actually meant. The final step toward effective understanding is to decide what to do as a results. One possible thing to do is to record what was said as a fact and be done with it. For some sentences, whose intended effect is clearly declarative, that is precisely correct thing to do. But for other sentences, including this one, the intended effect is different. We can discover this intended effect by applying a set of rules that characterize cooperative dialogues. The final step in pragmatic processing is to translate, from the knowledge based representation to a command to be executed by the system.

STEPS INVOLVED IN PREPROCESSING

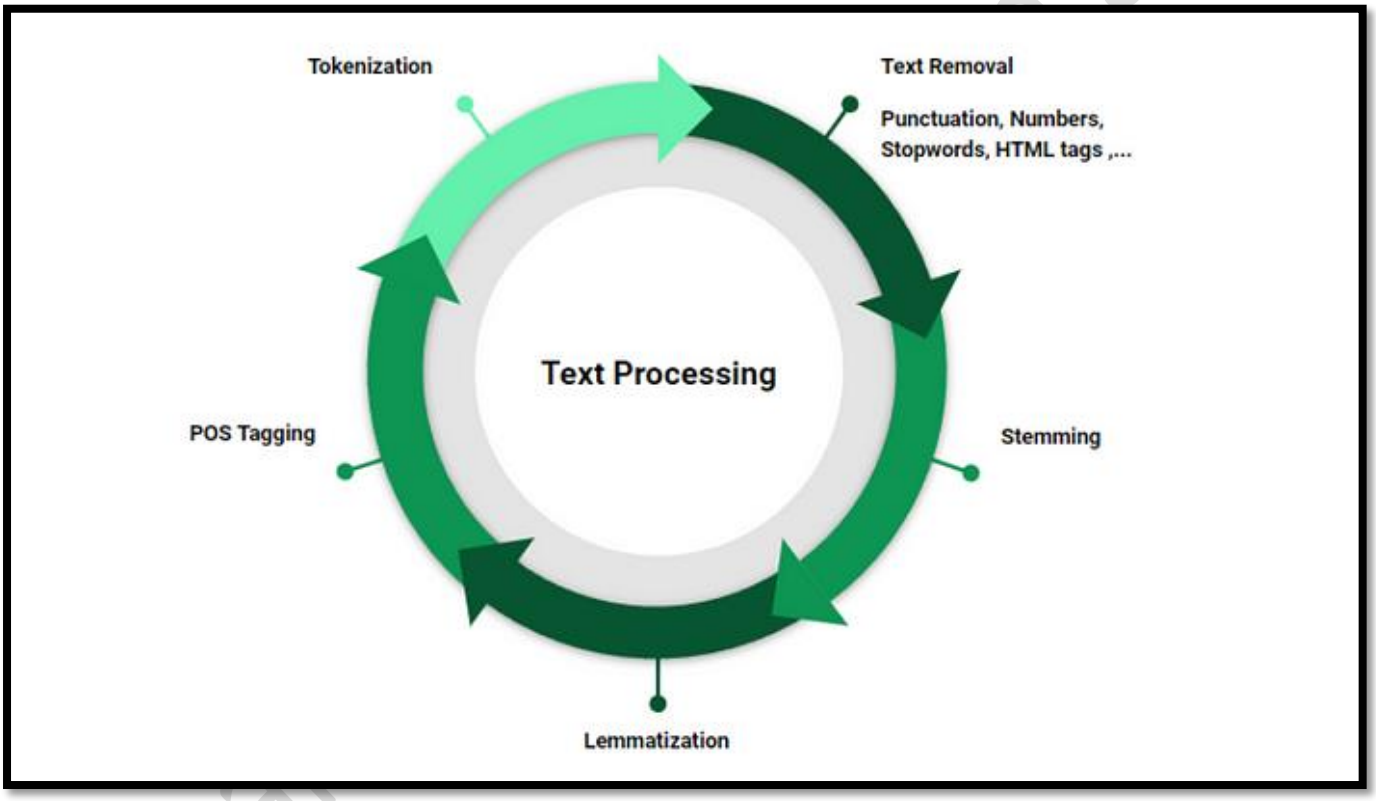


Figure: Preprocessing steps

Segmentation

Segmentation mainly refers to sentence segmentation or sentence tokenization wherein the continuous dataset or a block of text is broken down into meaningful sentences to ease the process of extracting features from text. The breaking point or the point of tokenization for a sentence would commonly be a full stop or a comma in some cases. The text is broken into another one when a full stop or any significant punctuation mark is encountered which makes the remaining part of the sentence look meaningful. Segmentation isn't always carried out for every kind of use case. It depends on the size of every text in the corpus which determines the necessity to use this technique.

Removal of punctuations, special characters and URLs

Raw data has lots of instances of punctuations or special characters (@, $, *,) which are not of much importance nor is understood by the machine. Therefore, its existence in data just contributes to the noise in it and should be removed. This is done by using regular expressions to eliminate all kinds of punctuations and special characters that are encountered. Regular expressions are also used to get rid of URLs in the text that aren't important usually.

Lowercasing

After successful removal of punctuations and URLs, the text must be converted to lowercase. Lowercasing the text is important as the machine might consider the same word again if it is written in uppercase or any other manner. For example the word 'love' and Love' might be considered differently and would be assigned different vectors when the text is vectorized for feature extraction. This step might be neglected at times, but it is the quite efficient and the simplest method in preprocessing which enhances the possibility of accurate results.

Tokenization

Unlike sentence tokenization, this step primarily breaks up or splits the sentence into an array of words which are referred to as tokens. The sentence is usually split up on space between two words or even when a punctuation is encountered depending on what condition might be applied. For example, the sentence: Text preprocessing is an important step in NLP. After tokenization will appear as: "Text", "preprocessing", "is", "an", "important", "step", "in", "NLP", "."

Parts-of-Speech Tagging

To understand the semantic meaning of the text, POS tagging proves to be useful. POS tagging is the classification of text into the various parts of speech such as noun, adjective, verbs, preposition and so on. POS tagging can be done after sentence segmentation as well so that the correct context of the sentence is understood, and the words are tagged in a better way. This technique should be applied before Stop words removal for the provision of accurate results. It can also be considered to apply this technique before removing the punctuations as well for more accurate results. If POS tagging has to be carried out, then these factors should definitely be considered.

Removing Stop words

NLP is extracting keywords about a particular topic depending on the use case. Hence for text classification or other problems, word such as 'the', 'a', 'is', 'are', 'an', etc. are not of importance and are quite often discarded. Such words are known as stop words and need to be identified as efficiently as possible. Tokenization of text helps in identifying such words easily without much hassle. This technique need not be applied always. The kind of problem must be considered before its application.

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Text Normalization

Text Normalization can prove to be very useful when analysing social media comments. It is used to standardize the text which helps in the elimination of noise that is contributed to the data when people usually express themselves on social media.

Stemming

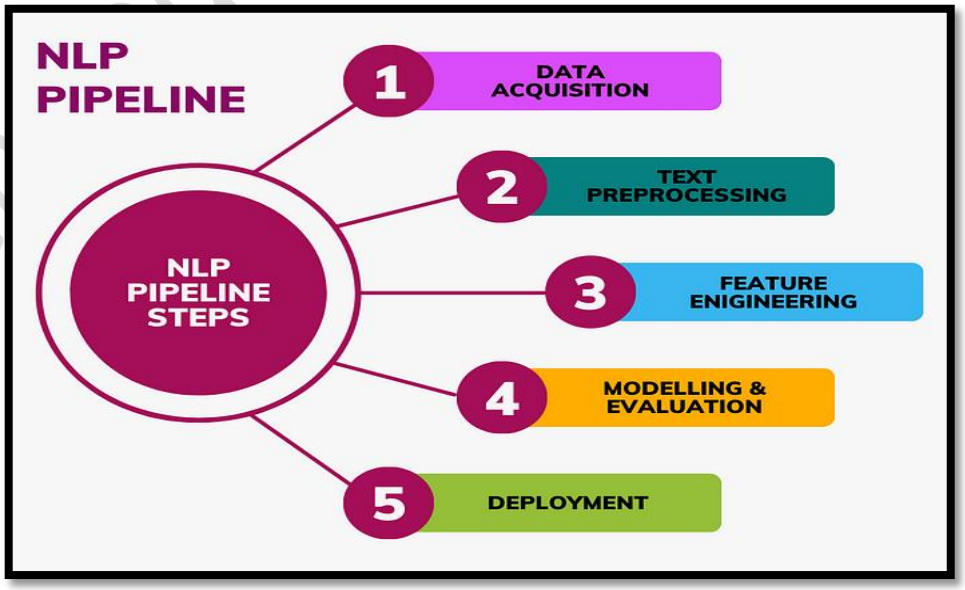
Stemming is a technique used to shorten the word and bring it to its root form. The technique results in shortening the word to an extent where the semantics are preserved but the meaning is lost in some cases. For example, the word 'connect' will be stemmed as 'connect' but the word 'trouble' will be stemmed as 'trouble'. The problem should be well studied and then it should be decided whether to use stemming for preprocessing. Stemming might prove useful in google searches as all possible searches related to the typed words need to be shown. Porter Stemmer is the commonly used algorithm for stemming.

Lemmatization

Lemmatization is quite like stemming but the meaning of the word is very well preserved with it. The word that is transformed to its root form is known as lemma which preserves the semantics as well the meaning of the word. An example for how lemmatization is different from stemming would be that the words 'bullying' and 'bullied' would be stemmed as 'bulli' but it would be lemmatized as 'bully'. Lemmatization in at times preferred over stemming because the meaning of the word is preserved.

NATURAL LANGUAGE PROCESSING PIPELINE AND ITS PROBLEMS

In Natural Language Processing (NLP), an NLP pipeline is a sequence of interconnected steps that systematically transform raw text data into a desired output suitable for further analysis or application. It's analogous to a factory assembly line, where each step refines the material until it reaches its final form.



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Here's a breakdown of the common stages in an NLP pipeline:

1. Data Acquisition
2. Text Preprocessing
3. Feature Engineering
4. Modelling
5. Evaluation
6. Deployment

Points to remember:

- This pipeline is not universal.
- This is ML pipeline and deep learning pipelines are slightly different.
- NLP pipeline is non-linear (that means stages can have more dynamic connections, allowing for branching and iteration).

Now, let's discuss the different NLP pipeline steps one by one in detail:

1. Data Acquisition

Data acquisition involves obtaining raw textual data from various sources to create a robust dataset for NLP tasks. It involves assessing the availability and accessibility of data, whether it's readily available, needs supplementation, or requires creation from scratch.

While data acquisition, you will always face one of three situations:

(i) Data Available Scenarios

Here, you can further have three situations.

1. **Data on Your Desk:** The data needed for the NLP task is already in possession. Initiate the text preprocessing stage immediately.
2. **Data in Databases:** The required data resides within company databases or repositories. Collaborate with data engineers to retrieve the data.
3. **Less Data:** Insufficient data volume for robust model training or analysis. Employ data augmentation techniques to enhance the dataset.

Data Augmentation Techniques:

- **Synonym Replacement:** Replace words with their synonyms to diversify the dataset without altering the context significantly.
- **Bigram Flip:** Alter word sequences by flipping bigrams to create variations.
- **Back Translation:** Translate text to another language and then back to the original language, introducing diverse phrasing.
- **Adding Noise:** Introduce random noise or perturbations to augment data.

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(ii) Data from Other Resources

In scenarios where data isn't readily available or needs supplementation, several strategies come into play:

- **Public Datasets:** Utilize publicly available datasets from repositories like Kaggle, UCI Machine Learning Repository, or government databases, aligning them with the project's requirements.
- **Web Scraping:** Extract data from websites or forums by scraping relevant information. Tools like BeautifulSoup or Scrapy assist in collecting data from various websites.
- **APIs:** Access data through Application Programming Interfaces (APIs) offered by various platforms such as social media APIs (Twitter/X, Reddit), news aggregators, or linguistic databases.
- **PDFs:** Extract text from PDF documents relevant to the project using libraries like PyPDF2 or PDFMiner.

(iii) Nobody has the Data

In cases where data isn't available through conventional means, organizations might resort to alternate strategies:

- **Engaging Trusted Clients:** Collaborate with trustworthy clients or users willing to share anonymized data relevant to the project's goals.
- **Data Generation:** If viable, companies can generate synthetic data or collect information through surveys, interviews, or user-generated content to build a dataset from scratch.

2. Text Preprocessing:

Text preprocessing is a critical phase in NLP, encompassing various steps to refine raw text data for meaningful analysis and model training. Let's delve deeper into each stage:

(i) Basic Cleaning:

This initial stage focuses on eliminating irrelevant or disruptive elements from the text:

- **HTML Tag Removal:** Stripping out HTML tags is crucial when working with web-based text sources. These tags contain formatting information and are unnecessary for linguistic analysis.
- **Handling Emojis:** Managing emojis involves converting them to textual representations using emoji library or removing them entirely, depending on their relevance to the analysis.
- **Basic Spell Checks:** Performing rudimentary spell checks to rectify common typographical errors and ensure consistency in the text.

(ii) Basic Preprocessing

Here, the primary goal is to prepare the text for further analysis by applying fundamental transformations:

- **Tokenization:** Segmenting text into smaller units such as words or sentences (word tokenization and sentence tokenization). This step breaks down the text into manageable chunks.
- **Stop Word Removal:** Eliminating common and less meaningful words (stop words) like "the," "is," etc., which don't contribute significantly to the meaning of the text.

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- **Stemming/Lemmatization:** Reducing words to their root forms — stemming removes prefixes/suffixes, while lemmatization maps words to their base or dictionary form, aiding in standardization.
- **Lowercasing:** Converting all text to lowercase to ensure uniformity in text analysis, as case sensitivity can affect certain NLP tasks.
- **Language Detection:** Identifying the language of the text, is especially useful when dealing with multilingual content.

(iii) Advanced Preprocessing:

This stage involves more intricate linguistic analysis to delve deeper into the structural and semantic aspects of the text:

- **Part-of-Speech (POS) Tagging:** Assigning grammatical categories (like nouns, verbs, adjectives) to words in the text, providing insights into the syntactic structure.
- **Parsing:** Analyzing the grammatical structure of sentences to identify relationships between words and determining the syntactic roles and dependencies.
- **Coreference Resolution:** Resolving references within the text, linking pronouns or noun phrases to their respective entities for coherent understanding and analysis.

3. Feature Engineering

Feature engineering in Natural Language Processing (NLP) involves transforming raw text data into numerical features that machine learning models can comprehend and utilize effectively. The goal is to represent text in a format that captures semantic meaning, contextual information, and relationships between words.

It can be done through various techniques:

(i) Bag of Words (BoW)

- Represents text as a collection of unique words disregarding grammar or word order.
- Creates a matrix where rows represent documents and columns represent unique words, with values indicating word occurrence frequencies.
- Simple yet effective, but loses sequence information and context.

(ii) Term Frequency-Inverse Document Frequency (TF-IDF)

- Measures the importance of words in a document relative to a corpus.
- Considers both the frequency of a term in a document and its rarity across the corpus.
- Assigns higher weights to rare terms that are more discriminative.

(iii) One-Hot Encoding

- Represents words as binary vectors, where each word has a unique index in the vector.
- Converts words into a high-dimensional space, with each dimension corresponding to a unique word.
- Effective for small vocabularies but leads to high dimensionality and sparsity issues for large datasets.

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(iv) Word Embeddings (Word2Vec, GloVe, FastText)

- Techniques that map words or phrases to dense vector representations in a continuous vector space.
- Capture semantic relationships between words by placing similar words closer in the vector space.
- Retain semantic meaning and context, useful for capturing word analogies and semantic similarities.

(v) N-Gram Models

- Captures sequences of adjacent words (bigrams, trigrams, etc.) as features.
- Preserves some sequence information, aiding in capturing context in language.

(vi) Dependency Parsing

- Represents the grammatical structure of sentences as features.
- Captures relationships between words through syntactic dependencies.

In **ML-based applications**, data scientists actively engineer features leveraging domain expertise to handcraft relevant inputs for the models, aligning with the problem domain.

4. Modelling

The heart of the pipeline, where models are applied and evaluated using different approaches:

(i) Heuristic Approaches

Heuristic models rely on predefined rules or strategies based on expert knowledge to make decisions.

Application: Commonly used in simple text-based tasks where rule-based systems can effectively handle specific patterns or tasks, like keyword matching for sentiment analysis or rule-based chatbots.

(ii) Machine Learning (ML) Approaches

ML models learn patterns and relationships from data to make predictions or classifications.

Applications:

- **Support Vector Machines (SVM):** Effective for text classification tasks by finding the best separation between classes in a high-dimensional space.
- **Random Forests:** Suitable for tasks like sentiment analysis or text categorization, leveraging ensemble learning for improved accuracy.

(iii) Deep Learning (DL) Approaches

DL models use neural networks with multiple layers to learn complex patterns and representations from raw data.

Applications:

- **Recurrent Neural Networks (RNNs):** Effective for sequence-based tasks like language modelling, sentiment analysis, or machine translation.

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- **Transformers:** Particularly potent for attention-based mechanisms, excelling in tasks like language translation, text generation, and summarization due to their ability to capture long-range dependencies efficiently.

(iv) Cloud APIs

Cloud-based APIs offer pre-built, scalable models accessible via APIs, saving time and resources.

Applications:

- **Google Cloud Natural Language API:** Offers sentiment analysis, entity recognition, and language detection.
- **Microsoft Azure Text Analytics API:** Provides sentiment analysis, key phrase extraction, and named entity recognition.

Selection Criteria:

- **Problem Domain:** Different tasks require different model architectures. Heuristic methods suit specific, rule-based tasks, while ML and DL excel in learning from data.
- **Data Volume and Complexity:** ML and DL approaches often require substantial data volumes for effective learning, with DL being more data-hungry for complex tasks.
- **Resource Availability:** Cloud APIs are convenient for quick prototyping or when resources for training and maintaining models are limited.

5. Evaluation

Evaluation in the NLP pipeline is pivotal, encompassing intrinsic and extrinsic assessments to comprehensively gauge model performance from both technical and practical standpoints.

(i) Intrinsic Evaluation

Intrinsic evaluation focuses on assessing the technical aspects and capabilities of the model in isolation, without considering its real-world application.

Examples of Intrinsic Metrics:

- **Accuracy:** Measures the ratio of correctly predicted instances to the total instances in the dataset.
- **Precision, Recall, F1-score:** Assess the model's performance in binary or multi-class classification tasks.
- **Perplexity:** Evaluates the language model's predictive capability in language generation tasks.
- **BLEU Score:** Measures the quality of machine-translated text against a reference translation.

(ii) Extrinsic Evaluation

Extrinsic evaluation measures the model's performance in real-world applications or business contexts, considering its impact and utility in practical scenarios.

Examples of Extrinsic Evaluation Metrics:

- **Business Metrics:** Metrics aligned with specific business goals or outcomes, such as customer satisfaction scores, revenue impact, or user engagement rates.

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- **Task-Specific Metrics:** Metrics directly relevant to the NLP task at hand, like sentiment analysis accuracy for customer feedback or document classification precision for information retrieval systems.
- **User-Centric Evaluation:** Soliciting user feedback, surveys, or usability testing to assess user satisfaction and experience with the NLP application.

Importance of Intrinsic and Extrinsic Evaluation:

- **Technical Assessment (Intrinsic):** Intrinsic metrics provide insights into the model's performance on specific tasks or benchmarks, helping fine-tune model parameters and architectures.
- **Real-world Applicability (Extrinsic):** Extrinsic evaluation ensures that the model's performance aligns with practical requirements, determining its effectiveness and impact in real-world settings.

6. Deployment

The deployment phase in the NLP pipeline marks the transition of the developed model from the development environment to a production environment, followed by continuous monitoring and adaptation to ensure sustained performance and relevance.

(i) Deployment

- **Rolling out the Model:** Moving the trained NLP model from the development environment to a production environment where it can be utilized in real-world applications.
- **Infrastructure Setup:** Configuring the necessary infrastructure, integrating the model into the existing systems, and ensuring scalability and reliability.
- **Testing and Validation:** Thoroughly testing the deployed model to ensure it functions as expected and delivers accurate results in the production environment.

(ii) Monitoring

- **Continuous Performance Oversight:** Constantly monitoring the model's performance, including its accuracy, efficiency, and response time in real-time or at regular intervals.
- **Alert Systems:** Implementing alert systems or triggers to notify about deviations or anomalies in the model's behaviour, ensuring timely interventions.

(iii) Update

- **Adaptation to Dynamic Data:** Adapting the model to changing data patterns or evolving requirements by periodically updating and retraining the model.
- **Improvement Iterations:** Incorporating feedback, identifying areas for improvement, and fine-tuning the model to enhance its performance or address changing user needs.
- **Version Control:** Maintaining version control to track model iterations and changes, ensuring transparency and reproducibility.

Challenges and Considerations:

- **Data Drift:** Evolving data patterns might lead to data drift, impacting model performance. Regular updates help mitigate this challenge.

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- **Ethical and Legal Compliance:** Models must comply with ethical standards and legal regulations, especially when handling sensitive data or influencing critical decisions.
- **Resource Management:** Effective deployment requires resource management to ensure optimal utilization of computational resources and cost efficiency.

CHALLENGES IN NATURAL LANGUAGE PROCESSING

1. Misspellings

Natural languages are full of misspellings, typos, and inconsistencies in style. For example, the word "process" can be spelled as either "process" or "processing." The problem is compounded when you add accents or other characters that are not in your dictionary.

2. Language Differences

An English speaker might say, "I'm going to work tomorrow morning," while an Italian speaker would say, "Domani Mattina Vado al lavoro." Even though these two sentences mean the same thing, NLP won't understand the latter unless you translate it into English first.

3. Innate Biases

Natural processing languages are based on human logic and data sets. In some situations, NLP systems may carry out the biases of their programmers or the data sets they use. It can also sometimes interpret the context differently due to innate biases, leading to inaccurate results.

4. Words with Multiple Meanings

NLP assumes that language is precise and unambiguous. Language is neither precise nor unambiguous. Many words have multiple meanings and can be used in different ways. For example, when we say "bark," it can either be dog bark or tree bark.

5. Uncertainty and False Positives

False positives occur when the NLP detects a term that should be understandable but can't be replied to properly. The goal is to create an NLP system that can identify its limitations and clear up confusion by using questions or hints.

6. Training Data

One of the biggest challenges with natural processing language is inaccurate training data. The more training data you have, the better your results will be. If you give the system incorrect or biased data, it will either learn the wrong things or learn inefficiently.

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CONTENT BEYOND SYLLABUS

1	Levels of Natural Language Processing.
2	Difference between NLU and NLG.

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QUESTION BANK FOR UNIT 1

2 MARKS QUESTION/SHORT QUESTIONS

1	Define Natural Language Processing (NLP) and name two types of NLP systems.
2	List four key terminologies used in NLP and their meanings.
3	Explain how a computer understands text in NLP systems.
4	Identify two real-world applications of NLP and briefly explain how they use text processing.
5	Compare the roles of syntactic and semantic analysis in NLP systems.
6	Break down the process of how a computer converts raw text into meaningful data.
7	Assess the suitability of rule-based NLP systems for processing unstructured text data.
8	Justify why understanding context is critical for NLP systems in sentiment analysis.
9	Given a sentence, illustrate how tokenization and lemmatization are applied in NLP.
10	Describe the difference between rule-based and statistical NLP systems.
11	Propose a simple NLP system to categorize customer reviews as positive or negative.
12	Design a basic workflow for an NLP system to extract names from a text document.
13	List the main steps involved in the NLP pipeline.
14	Name four common text pre-processing techniques used in NLP.
15	Explain why text pre-processing is essential before applying NLP algorithms.
16	Describe two major challenges faced in NLP and their impact on system performance.
17	Demonstrate how stop-word removal and stemming are used in pre-processing a given text.
18	Apply the concept of tokenization to break down a sample sentence for NLP processing.
19	List levels of NLP.
20	State the difference between NLU and NLG.

6/7/8 MARKS QUESTION/ LONG QUESTIONS

1	Define Natural Language Processing (NLP) and discuss its significance in modern technology. List and briefly describe four types of NLP systems and four key terminologies used in NLP, providing examples for each terminology.
2	Explain in detail how a computer understands and processes text in NLP systems. Include the role of tokenization, part-of-speech tagging, and semantic analysis, and illustrate with a sample sentence.
3	Demonstrate how an NLP system can be used to develop a chatbot for customer support. Describe the key components involved, such as text parsing and intent recognition, and provide a step-by-step process for handling a sample customer query.

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4	Evaluate the effectiveness of current NLP systems in understanding contextual nuances in human language, such as sarcasm or idioms. Discuss the limitations and suggest two improvements to enhance context understanding.
5	Design an NLP system to extract and summarize key information from news articles. Outline the architecture, including input processing, key components, and output format, and explain how it handles diverse text structures.
6	Describe the complete NLP pipeline, listing and explaining each step involved, from text input to final output. Include four pre-processing techniques and their purposes in preparing text for analysis.
7	Explain the importance of text pre-processing in NLP and discuss four key pre-processing steps (e.g., tokenization, normalization, stop-word removal, stemming). Provide examples to show how each step impacts the quality of NLP outcomes.
8	Apply the NLP pipeline to process a sample paragraph from a social media post to perform sentiment analysis. Describe each step, including pre-processing, feature extraction, and classification, and show how the pipeline leads to a sentiment score.
9	Analyze the challenges of ambiguity and data sparsity in NLP systems. Discuss their causes, impacts on performance, and two techniques to mitigate each challenge, providing examples to illustrate your points.
10	Critically evaluate the impact of poor pre-processing on the performance of an NLP model. Discuss how issues like incomplete stop-word removal or improper tokenization affect accuracy, and propose strategies to address these issues.
11	Design a pre-processing pipeline for an NLP system to analyse multilingual customer feedback. Specify the steps to handle multiple languages, address challenges like code-switching, and ensure accurate sentiment analysis, including tools or techniques for each step.
12	Discuss how computers process human language in NLP systems, focusing on the roles of syntactic parsing and named entity recognition. Use a sample sentence to demonstrate how these processes contribute to text understanding.
13	Assess the challenges of achieving human-like language understanding in NLP systems, particularly in handling ambiguous phrases. Discuss two limitations and propose methods to improve ambiguity resolution.
14	Enumerate and describe the core steps in the NLP pipeline, from data acquisition to model output. Explain how three pre-processing techniques (e.g., lemmatization, tokenization, vectorization) fit into the pipeline and their significance.
15	Apply the NLP pipeline to develop a system for detecting spam emails. Describe each step, including pre-processing, feature extraction, and classification, and show how a sample email is processed to determine if it is spam.
16	List and describe five key pre-processing techniques used in NLP, such as tokenization, normalization, and stop-word removal. Explain their purpose and provide an example of how each technique transforms a sample sentence.

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